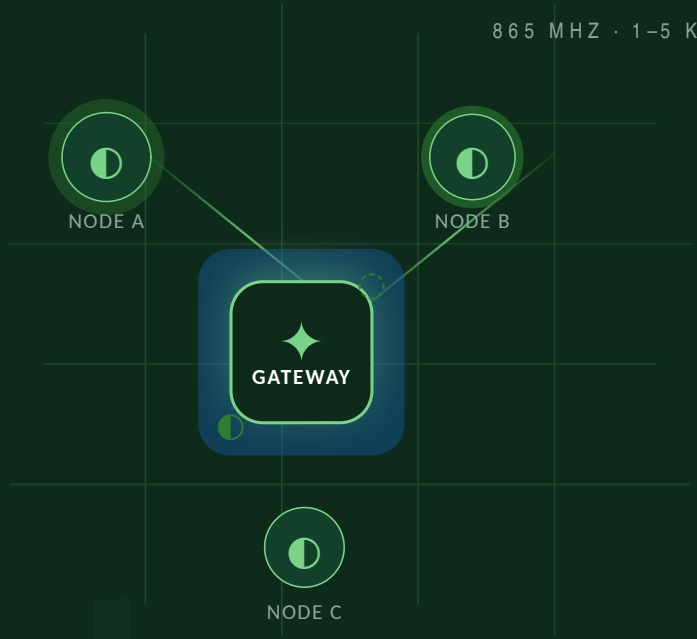
UTKARSH 1.0 · IIT BOMBAY × HONEYWELL · 2026

LORA · EDGE AI · CREDIT

A PADDY-FIELD INTELLIGENCE SYSTEM

AGRI-GRID



LoRa Sensor Grid for Paddy Water & Methane Intelligence

LoRa Grid · Edge AI · Credit Engine

TEAM [TBD] · PROBLEM STATEMENT ID [TBD] · MENTOR [TBD]

SUSTAINABILITY · ESG · MRV

THE PROBLEM

Paddy farming is drowning in water — and methane

30–50% Avoidable irrigation water wasted in continuously flooded paddy ◐

~ 10–12% Of agricultural greenhouse-gas emissions come from paddy methane ☁

Income loss Over-irrigation raises input cost and cuts farmer margin per hectare ↘

flooded fields · 30–50% avoidable water · methane hotspot



Flooded paddy, monsoon India

WHY CURRENT TOOLS FAIL

Single sensors. No AWD. No proof.

THREE FAILURE MODES

01 / IOT KIT

Generic IoT kit

Single WiFi node, 10–50m range, one point measurement. No spatial view of the field.

MISSING: SPATIAL GRID · LONG RANGE

02 / MANUAL

Manual observation

Farmer walks paddy zones daily to eyeball water level. Misses anomalies overnight, no audit record.

MISSING: CONTINUOUS DATA · PROOF

03 / APP-ONLY

App-only dashboards

Pretty UI on top of sparse or simulated data. No AWD cycle, no methane math, no credit eligibility.

MISSING: AWD · CREDIT ENGINE

point-only · no spatial · no credit

OUR SOLUTION

A coordinated sensor grid that turns water data into credit

3 PILLARS · 1 BUILD

PILLAR 01

LoRa Grid

Spatial sensing across paddy zones

- ✓ 865MHz IN865 · 1–5 km range
- ✓ 3+ nodes per field
- ✓ Star topology → 1 gateway

SENSE

PILLAR 02

Edge AI

On-device anomaly detection

- ✓ IsolationForest (sklearn)
- ✓ AWD cycle detection
- ✓ Explainable scores 0–1

IDENTIFY

PILLAR 03

Credit Engine

AWD → methane → water/carbon credit

- ✓ Days-in-AWD compliance
- ✓ Methane −30–50%
- ✓ MRV-ready eligibility flag

DECIDE

SYSTEM ARCHITECTURE

Star grid → cloud → Agri Stack → dashboard

✦ 4 LAYERS

LAYER 04 — DASHBOARD + CREDIT
Web heatmap dashboard · AWD→methane→credit engine · Alert panel · Agri Stack adapter

↑ HTTPS · REST

LAYER 03 — CLOUD
FastAPI / ThingsBoard · PostgreSQL + PostGIS · IsolationForest anomaly model · UFSI schema adapter (mock)

↑ WIFI/HTTP ↑

LAYER 02 — GATEWAY
Heltec WiFi LoRa 32 V3 · ESP32 + SX1262 · WiFi uplink · OLED debug display

↑ LORA 865MHZ IN865 ↑

LAYER 01 — FIELD NODES × 3
Node A (NW edge) · Node B (NE edge) · Node C (S center) — each: water level + temp + soil moisture + battery

DATA FLOW ↑

- Sensor data
- Aggregated + identified
- Credit-decisioned

⚡ OFFLINE-FIRST

Gateway buffers LoRa frames if WiFi drops; replays on reconnect.

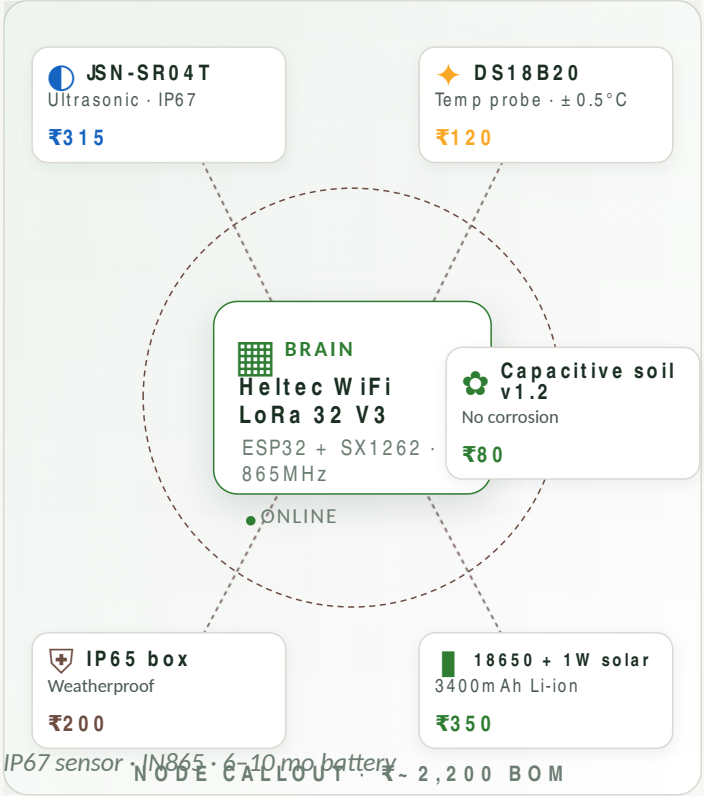
TOPOLOGY

✦ Star · 3 nodes → 1 gateway

865MHz IN865 · star topology · offline-first

One node: ₹~2,200 · months of solar life

5 SENSORS · 1 BRAIN



Heltec WiFi LoRa 32 V3	ESP32 + SX1262 + OLED, 865MHz IN865
JSN-SR04T	Waterproof ultrasonic, IP67, ±3mm
DS18B20	Waterproof temp probe, -55 to +125°C
Capacitive soil v1.2	No corrosion, air=0% / sat=100%
18650 + 1W solar	3400mAh, 6–10 months deep-sleep
IP65 box	Weatherproof field enclosure

HONEST NOTE

6–10 months battery-only with true deep sleep. Heltec V3 quiescent (~100µA+) cuts this to 1–3 months — cut board power with a MOSFET between reads to hit the full claim.

IP67 sensor · IN865 · 6–10 mo battery

NODE CALLOUT · ₹~2,200 BOM

HOW IT WORKS

From field to farmer credit in 4 hops

01

SENSE

3 nodes measure water level, temp, soil moisture every 5 min. Deep-sleep between reads.

5-MIN CADENCE

→

02

EDGE-AI

On-device IsolationForest flags anomalies + AWD cycle detection. Score 0–1 per reading.

NO CLOUD ROUND-TRIP

→

03

CLOUD

Gateway forwards to FastAPI. Zone data linked to FPO farmer ID via UFSI schema (mock).

UFSI-LINKED

→

04

CREDIT

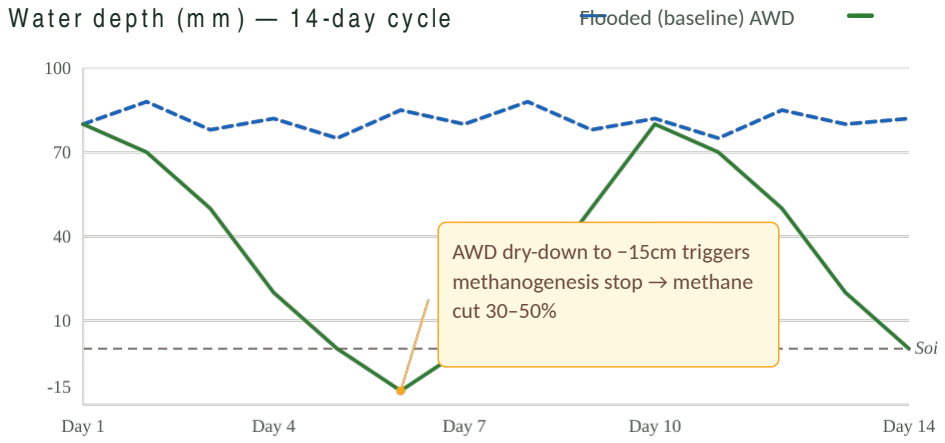
Dashboard computes methane reduction (kg CO_2e/ha) + water/carbon credit eligibility flag.

MONETIZABLE

sense → infer → link → credit

THE DIFFERENTIATOR

Alternate Wetting & Drying cuts methane 30–50%



30–50%

CH_4 REDUCED

methane cut vs continuous flooding

Source: IRRI AWD field trials

72%

CREDIT ELIGIBILITY

driven by AWD adherence + water saved

MRV-ready output · per zone

AWD · – 30–50% CH_4 · credit-ready

AGRI STACK INTEGRATION

Agri-Stack-ready: farmer identity, sandbox today

SENSOR ZONE

Each paddy zone maps to a geo-referenced plot boundary from Agri Stack village maps.

GEO-REFERENCED PLOT

→

FPO FARMER ID (UFSI MOCK)

Zone linked to farmer identity via UFSI schema — simulated Farmer Registry + Crop Sown payloads.

OPENAPI 3.0 SCHEMA

→

SCHEME ELIGIBILITY

PM-KUSUM / PMFBY eligibility flag computed from AWD compliance + plot data.

AUTO-FLAGGED PER CYCLE

⚠ **HONEST SCOPE**

Live UFSI / Agri Stack API is **not** openly callable by students — it requires developer-portal registration, MoAFW use-case review, sandbox credentials, and CERT-In empaneled security audit for production. We build against the published OpenAPI 3.0 UFSI schema with simulated payloads and ship a clean adapter module — when sandbox credentials arrive, live calls drop in with **zero code change**. We claim *"Agri-Stack-ready (sandbox-integrated)"*, not "live in prod".

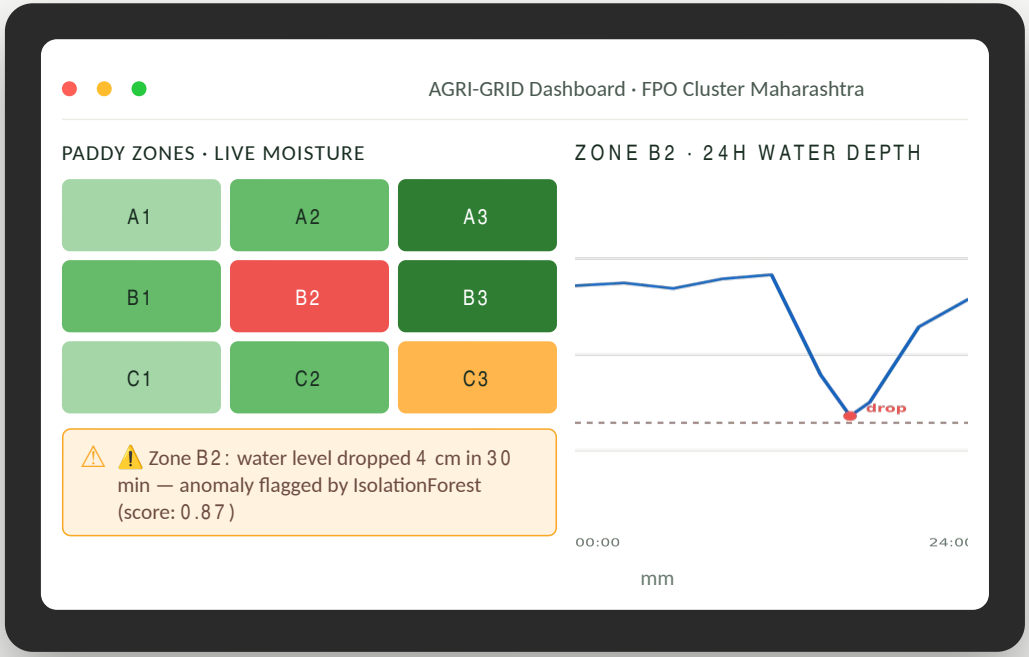
DASHBOARD & ALERTS

One field. Live spatial view.

LIVE KPI STRIP

Water saved (m^3)	1,240
Methane cut ($kg\ CO_2e/ha$)	180
Credit eligibility	72%

Live KPIs update every 5 min from grid



heatmap · alert · live

AI / ML COMPONENT

Edge anomaly detection, not just dashboards

PIPELINE · ON-DEVICE

1 **INPUT: 4-NODE TIME SERIES**

water level · temp · soil · battery (5-min cadence)

↓

2 **MODEL: ISOLATIONFOREST (SKLEARN)**

on-device, no cloud round-trip

↓

3 **SCORE: 0–1 PER READING**

threshold > 0.7 = anomaly

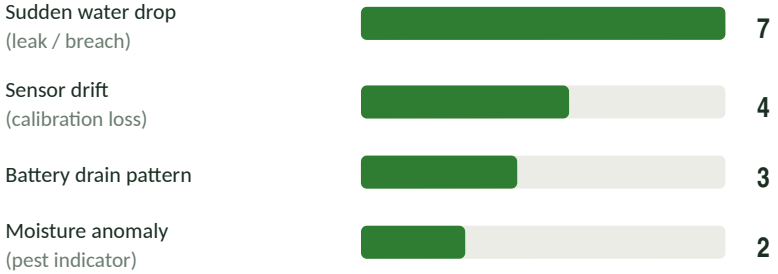
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4 **OUTPUT: ALERT + AWD CYCLE TAG**

explains which feature drove the score

Anomalies caught vs baseline (30-day window)

Count of confirmed anomaly events



Baseline (simple threshold) — same 30 days



Baseline = simple threshold alerts: catches only 2 of 7 leak events.

on-device · IsolationForest · explainable

COMPETITIVE LANDSCAPE

vs paddyAkvo & generic IoT

Feature	paddyAkvo (Enthu Tech)	Generic IoT kit	AGRI-GRID
Grid spatial view	single node	single node	3+ node grid ✓
Edge AI anomaly detection	no	threshold only	IsolationForest on-device ✓
Agri Stack identity (UFSI)	no	no	schema-ready, sandbox mock ✓
AWD → methane → credit engine	partial (methane claim only)	no	full (AWD + methane + credit) ✓
Cost (3-zone build)	proprietary / SaaS	₹5K–10K	₹14,995 ✓

paddyAkvo is the closest rival — we add multi-node grid + edge AI + Agri Stack FPO identity in one ₹15K build.

IMPACT & SDG ALIGNMENT

Measurable ESG impact

2 Zero Hunger

AWD sustains paddy yield with less water/input

6 Clean Water

Cuts irrigation water 15–30% per cycle

9 Industry & Innovation

Resilient rural LoRa IoT infrastructure

12 Responsible Consumption

Water/carbon-credit eligibility drives efficiency

13 Climate Action

AWD cuts paddy methane 30–50%

15–30%

irrigation water saved per paddy cycle (vs continuous flooding)

30–50%

methane (CH_4) reduction via AWD compliance

1 FPO

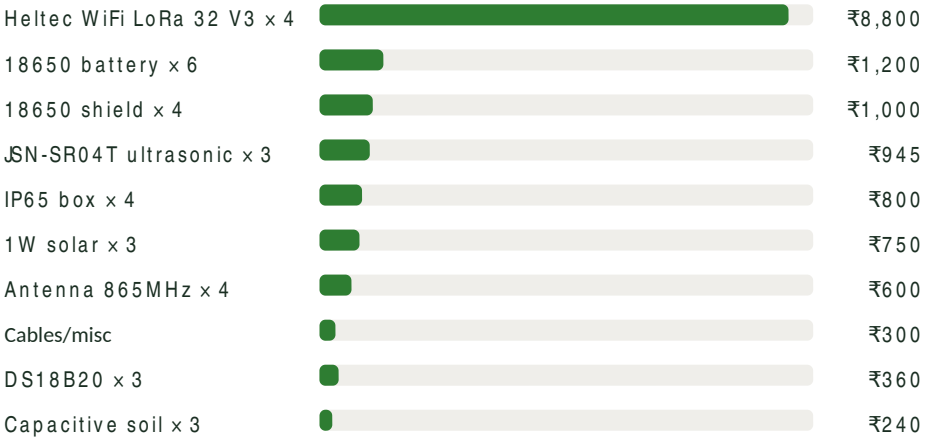
pilot cluster (10–50 farmers scalable to district)

SDG 2/6/9/12/13 · ESG · scalable

FEASIBILITY & COST

₹15K. 4 days. Honest risks.

Cost breakdown — total ₹14,995



TOTAL: ₹14,995 · within ₹15K budget

Lead-time (LoRa boards) — MED

Robocraze same-day shipping in Bangalore; order Day 1 morning.

LoRa range in paddy — MED

Star topology fallback if mesh fails; 1–5 km realistic at 865MHz IN865, +30 dBm EIRP.

Mentor sign-off — MED

Email Day 1; IIT-B / IHCFSI faculty targets identified.

Agri Stack API gating — MED

Schema-driven mock with simulated payloads; adapter ready for live calls when sandbox arrives.

₹14,995 · 4-day · risk-aware

BUILD ROADMAP

Built in 4 days, July 12–15

JUL 12 · JUL 13 · JUL 14 · JUL 15

DAY 1

JUL 12

Hardware + Firmware

Order components (Robocraze same-day)

Flash Heltec boards, wire sensors

Test LoRa link between 2 nodes

DAY 2

JUL 13

Backend + Integration

Set up FastAPI/ThingsBoard + PostgreSQL

Data ingestion + anomaly model

Connect gateway → backend

DAY 3

JUL 14

Dashboard + Demo Prep

Heatmap + alerts dashboard

Agri Stack adapter (mock)

End-to-end dry run + demo rehearsal

DAY 4

JUL 15

SUBMIT ✓

Final hardware test

Deploy dashboard to cloud

Submit on events-ihcfsi.org

Post-submission: pilot deployment at 1 FPO cluster (Maharashtra / Tamil Nadu) + credit onboarding with MRV partner.

DEMO PLAN

3-minute live demo

Story arc: problem → proof → impact

01

0:00–0:30

LIVE GRID READING

Open dashboard. Show 3 nodes streaming real water level + temperature + soil moisture from the on-stage paddy setup. LoRa packets arriving every 5 seconds.

PROOF:

real sensor data on screen, not mocked

02

0:30–1:30

DASHBOARD HEATMAP + ALERT

Pour water onto Zone B2 sensor live. Heatmap cell turns amber then red. Alert toast fires with IsolationForest anomaly score 0.87.

PROOF:

spatial view + edge AI working in real time

03

1:30–3:00

AWD → CREDIT NUMBER

Switch dashboard to AWD compliance view. Show methane reduction (kg CO_2e/ha) and credit eligibility % computed live from accumulated grid data.

PROOF:

the credit engine — our differentiator — runs end-to-end

live · spatial · credit

TEAM

Who builds it



Neeraj

AI / ML · BACKEND

Backend API (FastAPI)

Anomaly detection (sklearn IsolationForest)

Agri Stack UFSI adapter

PostgreSQL + PostGIS schema



Amudullah

HARDWARE · FIRMWARE

Flash firmware to all 4 Heltec boards

Wire sensors + stilling tubes

LoRa 865MHz link test

Solar + battery power budget



Riddhi

FRONTEND · DEMO

Dashboard (ThingsBoard / React)

Spatial heatmap viz

Alert UI + KPI strip

Demo slides + dry run

▽ **MENTOR (TBD)**

IIT-B / IHCFSI faculty — email Day 1

PENDING

3 members · mentor TBD

THE ASK

Grid the paddy. Bank the carbon.

A ₹15K LoRa grid that turns paddy water data into measurable methane reduction, water savings, and farmer credit eligibility — Agri-Stack-ready, edge-AI checked, SDG-aligned.

✓ Mentor sign-off

✓ Live demo Day 4

✓ Submit Jul 15 on events-ihcfsi.org

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